USP General Chapter <795>

Pharmaceutical Compounding — Nonsterile Preparations

A Training Module for the Compounding Academy

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Learning Objectives

By the end of this module, you will be able to:

Define Scope

Define a compounded nonsterile preparation (CNSP) and identify what falls within and outside the scope of <795>.

Designated Person

Describe the responsibilities of the designated person and the personnel training requirements under <795>.

Facility Standards

Apply facility, equipment, and component-handling standards in daily compounding practice.

Assign BUDs

Assign Beyond-Use Dates using water activity (aw) and Table 4 of <795>.

Distinguish Chapters

Distinguish <795> from <797> (sterile compounding) and <800> (hazardous drug handling).

<795> establishes the minimum standards every compounder must meet to protect patients.

What Is a Compounded Nonsterile Preparation?

> "Combining, admixing, diluting, pooling, reconstituting other than as provided in the manufacturer's labeling, or otherwise altering a drug product or bulk drug substance to create a nonsterile preparation."

Dosage forms covered under <795>:

Solid Oral

Tablets, capsules, powders, lozenges prepared for oral ingestion

Liquid Oral

Solutions, suspensions, syrups, and elixirs for oral use

Rectal & Vaginal

Suppositories, creams, gels, and enemas for rectal or vaginal routes

Topical

Creams, gels, ointments, and pastes applied to skin or mucous membranes

Nasal & Sinus

Sprays and drops for local nasal or sinus application only

Otic

Preparations for intact eardrum only; perforated TM routes to <797>

Hazardous drugs always layer in <800> — <795> sets the floor; <800> adds mandatory HD controls on top.

If you alter a drug outside its FDA-approved labeling for a single patient, you are compounding.

Practices Outside the Scope of <795>

→ **Reconstitution per Labeling**

Reconstitution strictly per the manufacturer's approved labeling is **not** compounding under <795>.

→ **Repackaging**

Repackaging of conventionally manufactured drug products — refer to USP General Chapter <1178> for applicable standards.

→ **Tablet Splitting**

Simply breaking or cutting a tablet into smaller portions does not constitute compounding under this chapter.

→ **Administration Preparation — 4-Hour Rule**

Preparing a single dose for a single patient when administration begins **within 4 hours** (e.g., crushing a tablet or opening a capsule to mix with food or liquid) is excluded. This exception is **patient-specific and time-limited**.

→ **Nonsterile Radiopharmaceuticals**

Nonsterile radiopharmaceuticals fall under USP General Chapter <825>, not <795>.

⚠ Routine pharmacy tasks can fall outside <795>, but the line is narrow — when in doubt, treat it as compounding.

Why <795> Exists: The Five Risks

Compounding errors that can cause patient harm or death. Every requirement in <795> maps back to one of the following five risks:

1. Microbial Contamination

Excessive microbial contamination of the finished preparation.

2. Strength Variability

Variability from intended strength of correct ingredients (e.g., ±10% of labeled strength).

3. Incompatibilities

Physical and chemical incompatibilities between active or inactive ingredients.

4. Contaminants

Introduction of chemical and physical contaminants during the compounding process.

5. Substandard Ingredients

Use of bulk drug substances or excipients of **inappropriate quality** without verified COAs.

⚠ Every requirement in <795> maps back to one of these five risks.

The Designated Person(s)

Every compounding facility must name one or more individuals accountable for performance and operations. A solo compounder is, by definition, the designated person.

Training & Competency

Oversee the training program and verify competency of all compounding personnel before independent practice.

Component Selection

Select components used in compounding; ensure all APIs and excipients meet quality specifications with supporting documentation.

Monitor & Correct

Monitor and observe compounding activities; take immediate corrective action for any deficient practices identified.

SOP Implementation

Ensure all SOPs are fully implemented, deviations are investigated, and follow-up actions are documented.

Storage Procedures

Establish and document handling and storage procedures for all components and finished CNSPs.

The designated person must be identified by name in the facility's SOPs.

⚠ Accountability is named, written, and personal — not assumed.

Personnel Training and Competency

Required Training Cadence

- Initial training and competency demonstration **before** independent compounding begins.
- Re-training and re-evaluation at least **every 12 months** thereafter.
- All training documented per Section 14 of <795>.

Training Procedure Must Cover

- Understanding the requirements of <795>.
- Reading and interpreting SDSs and COAs.
- Reading and following all SOPs related to compounding duties.

Core Competencies Required

- Hand hygiene and garbing procedures.
- Cleaning and sanitizing the compounding area and equipment.
- Handling and transporting components and CNSPs.
- Measuring and mixing techniques appropriate to the dosage form.
- Proper use of equipment and compounding devices.
- Documentation: MFR completion, CR creation, and log maintenance.

⚠ Competency is demonstrated, observed, and documented — not assumed from experience.

Personal Hygiene and Garbing

Self-Screen — Report to Designated Person

- Rashes, recent tattoos, or oozing sores on exposed skin.
- Conjunctivitis or active respiratory infection.

Remove Before Entering Compounding Area

- Outer garments: coats, hats, jackets, bandanas.
- Hand, wrist, and exposed jewelry; piercings that interfere with garbing or hygiene.
- Earbuds and headphones.

Hand Hygiene — Box 1 of <795>

- Wash hands with soap and water for at least **30 seconds**.
- Dry completely with **disposable towels**.
- Don gloves immediately after drying.

⚠ **Alcohol-based hand rub ALONE is NOT sufficient.**

Gloves — All Compounding

- Required for all compounding activities; inspect for defects before use.
- Replace immediately if torn or compromised.
- Wipe or replace gloves before starting a CNSP with different components.
- HD compounding: PPE per <800>.

⚠ 30 seconds at the sink, then gloves on — every time.

Buildings and Facilities

Compounding Area

A designated area, defined in SOPs, used **only for compounding** while compounding is in progress. Must be well lit, clean, sanitary, in good repair, and contain **no carpet**. Arranged to prevent mix-ups between components, labels, in-process materials, and finished CNSPs.

Storage Area — Temperature Monitoring

Monitor temperature **manually at least once daily** on open days, or continuously with a recording device. Calibrate or verify monitoring equipment per manufacturer or at least every **12 months**. Investigate all excursions; discard CNSPs or components if integrity is compromised. Store all items **off the floor**.

Water Sources

Hot and cold potable water with an accessible sink required. Use **Purified Water (<1231>)**, distilled, or reverse-osmosis water for rinsing equipment used in compounding. Do not use tap water for equipment rinsing.

⚠ A clean, dedicated, temperature-controlled space is the foundation of every CNSP.

Cleaning and Sanitizing — Minimum Frequency (Surfaces)

Table 1 of <795>: Minimum cleaning and sanitizing frequency for compounding area surfaces. If cleaning and sanitizing are performed as separate steps, **always clean first, then sanitize.** Document cleaning daily on compounding days.

Site	Minimum Frequency
Work surfaces	Beginning and end of each compounding shift; after spills; when contamination is suspected; between CNSPs with different components
Floors	Daily on compounding days; after spills; when contamination is suspected
Walls	When visibly soiled; after spills; when contamination is suspected
Ceilings	When visibly soiled; when contamination is suspected
Storage shelving	Every 3 months; after spills; when contamination is suspected

⚠ Work surfaces are cleaned twice per shift and between every component change.

Cleaning and Sanitizing — Minimum Frequency (Equipment)

Table 2 of <795>: Minimum cleaning frequency for compounding equipment and ventilated enclosures. Certification requirements are separate from — and in addition to — routine cleaning.

Equipment	Minimum Frequency
CVE (Containment Ventilated Enclosure)	Beginning and end of each compounding shift; after spills; horizontal work surface cleaned between CNSPs with different components
BSC (Biological Safety Cabinet)	Beginning and end of each compounding shift; after spills; horizontal work surface between CNSPs with different components; under the work surface at least monthly
Other devices and equipment	Before first use, then per manufacturer's recommendations; if no manufacturer guidance, between CNSPs with different components

⚠ **CVE and BSC must be certified at least every 12 months.** Certification is annual; cleaning is per shift.

Equipment Standards

1 Material Compatibility

Surfaces contacting components must **not** be reactive, additive, or sorptive — select materials appropriate to the CNSP formulation.

2 Inspection & Accuracy Verification

Inspect equipment before every use. Verify accuracy per manufacturer's schedule or at least every **12 months**, whichever is more frequent. Document all verifications.

3 Post-Compounding Cleaning

After each compounding session, clean thoroughly to prevent cross-contamination of subsequent preparations.

4 Airborne Particle Evaluation

Activities generating airborne particles (weighing APIs, mixing powders) must be evaluated for use of a **closed-system processing device**: CVE, BSC, or single-use containment glove bag.

5 Written Process Evaluation

Document the airborne-particle process evaluation and its outcome in the facility's SOPs — the evaluation itself is a required record.

⚠ If powder can become airborne, the evaluation — and its outcome — must be in writing.

Component Selection

APIs Must:

- Comply with the USP-NF monograph if one exists for that substance.
- Be accompanied by a **COA** with specifications and test results confirming identity and purity.
- In the United States, be manufactured by an **FDA-registered facility**.

Water in Formulations

- **Purified Water** or better quality (e.g., Sterile Water for Irrigation) is required whenever a formulation includes water as an ingredient.

All Other Components Should:

- Be accompanied by a COA verifying monograph compliance and any additional facility-specified requirements.
- Be manufactured by an FDA-registered facility when possible.

Designated Person's Responsibility

Component selection is exclusively the **designated person's** responsibility. No API may be used without documented quality verification in hand.

Selection → Documentation → Approval → Use. This sequence is non-negotiable.

No COA, no compounding — APIs require documented quality before they touch a balance.

Component Receipt and Handling

On Receipt — Document:

Date received, quantity received, supplier name, lot number, expiration date, and any in-house or third-party testing results. Every shipment receives a unique internal lot entry.

Components Without Vendor Expiration Date

Mark the **date of receipt** indelibly on each packaging system. **Do not use after 3 years from receipt.** Use a shorter date if the component is also used in sterile compounding (<797>) or is known to degrade faster under storage conditions.

Before Each Use — Verify:

Visually re-inspect every packaging system for breakage, looseness, or deviation in appearance or texture. Verify identity against the label and confirm storage conditions were met since receipt.

Reject & Handling Rules

Reject and segregate any component of unacceptable quality. Examine each lot for deterioration before every use. **Do NOT return unused material to the original container** — discard it to prevent contamination.

⚠ **Three-year limit on undated components** — and the clock starts at receipt.

Master Formulation Record (MFR)

A detailed recipe and procedure for each unique CNSP formulation. One MFR per unique formulation — it is the recipe of record. Any change to an MFR must be approved and documented per SOP before implementation.

Required Content — Part 1

- Name, strength or activity, and dosage form.
- Identities and amounts of all components; relevant characteristics (particle size, salt form, purity grade, solubility).
- Container closure system specification.
- Complete preparation instructions (equipment, supplies, step-by-step procedure).
- Physical description of the final CNSP (color, texture, appearance).

Required Content — Part 2

- BUD and storage requirements with reference source.
- Calculations to verify quantities and strength.
- Labeling requirements (e.g., "shake well before use").
- QC procedures (pH measurement, visual inspection) and expected results.
- Other process information needed for repeatability (pH adjustment targets, temperature controls).

⚠ One MFR per unique formulation — it is the recipe of record.

Compounding Record (CR)

A per-preparation log built from the MFR. Each CR must be reviewed for completeness before release; the reviewer's identity and date must be documented directly on the CR.

Required Content — Part 1

- Name, strength or activity, and dosage form.
- Date — or date and time — of preparation.
- Internal identifier (prescription, order, or lot number).
- Identification of personnel compounding AND personnel verifying the final CNSP.
- Each component: name, vendor/manufacturer, lot number, expiration date.
- Actual weight or measurement of each component used.

Required Content — Part 2

- Total quantity compounded.
- Assigned BUD and storage requirements.
- Calculations to verify quantities and strength.
- Physical description of the finished CNSP.
- Results of QC procedures performed (pH, visual inspection findings).
- MFR reference number linking back to the master recipe.

⚠ The CR makes every CNSP traceable to its components — essential for recalls.

Release Inspection and Labeling

Visual Inspection at Completion

- Confirm color, texture, and physical uniformity meet the MFR description.
- Check emulsions for phase separation.
- Inspect container closure integrity (leaks, cracks, seal integrity).
- Confirm the CNSP and its label match the CR and the prescription or order.
- If not released same day, **re-inspect immediately before release**.

Container Label Must Display

- Internal identifier (barcode, Rx, order, or lot number).
- Active ingredient(s) with amounts, activity, or concentration.
- Storage conditions if other than controlled room temperature.
- BUD and dosage form.
- Total amount or volume if not obvious from the container.

Dispensed Labeling Adds

- Route of administration.
- Statement that the preparation is compounded.
- Special handling instructions and required warning statements.
- Compounding facility name and contact information when leaving the facility or healthcare system.

A label that mismatches the Rx, the MFR, or the CR triggers an automatic reject — no exceptions.

⚠ A label that mismatches the Rx, the MFR, or the CR is an automatic reject.

Beyond-Use Dates (BUDs)

Definition

The date, or hour and date, after which a CNSP must **not be used, stored, or transported**. Calculated in hours, days, or months from the date or time of preparation. **BUD ≠ Expiration date** — expiration applies to manufactured products and bulk substances under prescribed storage conditions.

Parameters That Drive a BUD

- Chemical and physical stability of the API and added substances.
- Compatibility and degradation of the container closure system.
- Potential for microbial proliferation in the formulation matrix.
- Significant deviations from essential compounding steps.

Water Activity (aw) Concept

aw measures the **available water** that can support microbial growth and hydrolytic degradation reactions.

- aw is **NOT** the same as water content or percent moisture.
- Compounders are **not required to measure** aw directly.
- Use **Table 3 of <795>** as the reference for representative formulation types.

Refer to USP <1112> for the science of water activity measurement.

⚠ BUD is conservative by design — it protects the patient, not the inventory.

Default BUD Limits — Table 4

Table 4 of <795> applies when **no USP-NF compounded preparation monograph** exists and **no CNSP-specific stability data** are available. The water activity threshold of 0.60 determines which row applies.

Preparation Type	BUD (days)	Storage
Nonpreserved aqueous (aw ≥ 0.60)	14	Refrigerator
Preserved aqueous (aw ≥ 0.60)	35	CRT or Refrigerator
Nonaqueous oral liquids (aw < 0.60)	90	CRT or Refrigerator
Other nonaqueous dosage forms (aw < 0.60)	180	CRT or Refrigerator

⚠ BUD cannot exceed the shortest remaining expiration of any starting component. For CNSPs made from compounded components, BUD generally cannot exceed the shortest BUD of those components.

⚠ Memorize the four numbers — **14, 35, 90, 180** — and the water-activity line at **0.60**.

Water Activity (aw) at a Glance

Nonaqueous Examples — aw < 0.60

Hydrophilic petrolatum ointment	0.396
PEG suppository	0.374
Compressed tablet	0.465
Lip balm stick	0.181
Oil-filled capsule	0.468
PEG/propylene glycol oral solution	0.009

BUD range: 90–180 days (Table 4)

Aqueous Examples — aw ≥ 0.60

HPMC water-based gel	1.000
Oral suspension base	0.992
Nasal spray	0.991
Lotion (o/w emulsion)	0.986
Shampoo	0.976
Simple syrup	0.831

BUD range: 14–35 days (Table 4)

Aqueous CNSPs (aw ≥ 0.60) with BUDs within Table 4 should include a suitable antimicrobial preservative. If preservatives are contraindicated, refrigerate when feasible.

The 0.60 threshold is the line between "low microbial risk" and "needs preservation."

Extending a BUD Beyond Table 4

1. **USP-NF Monograph Ceiling**

   If a USP-NF compounded preparation monograph exists for the CNSP, **do not exceed the monograph's BUD** — the monograph is the controlling authority.

2. **Stability-Indicating Analytical Study**

   A stability study covering the API(s), the complete formulation, **AND** the actual container closure system may support an extended BUD — up to a maximum of 180 days.

3. **Aqueous CNSPs: Preservative Effectiveness**

   Aqueous CNSPs with extended BUDs must be tested for antimicrobial effectiveness per USP <51>. Passing AET is mandatory before the extended BUD is assigned.

4. **Acceptable AET Sources**

   In-house testing on the same formulation and container; results from an FDA-registered testing facility; peer-reviewed literature on the **same formulation and container closure system**.

5. **Bracketing Studies**

   Test high and low API concentrations to bracket intermediate strengths of the same formulation. All other ingredients — including preservative — must fall within the bracketed range to qualify.

⚠ **180 days is the ceiling — not a default.** An extended BUD requires real stability data plus, for aqueous, real preservative-effectiveness data.

SOPs, Quality Assurance, and Quality Control

Standard Operating Procedures

- Required for **every aspect** of the compounding operation — no activity lacks a governing SOP.
- All personnel trained on the SOPs that apply to their specific work duties.
- Designated person ensures full implementation and documented follow-up on all deviations.

QA Program Elements

- Adherence to compounding and documentation procedures.
- Prevention and detection of errors and quality problems.
- Evaluation of complaints and adverse events.
- Investigation and corrective action with documented outcomes.

QC Program Elements

- Sampling, testing, and documentation confirming specifications are met **before release** of each CNSP.
- pH measurement, visual inspection, weight verification, and other tests as defined in the MFR.

Annual Review Requirement

QA and QC programs are reviewed at least every 12 months by the designated person. All review results and any corrective actions taken must be documented.

> Quality is a written, reviewed, and actionable system — not a culture statement.

⚠ Quality is a written, reviewed, and actionable system — not a culture statement.

Complaints, Recalls, and Adverse Events

Recall SOP Must Address

- When to initiate — including immediate prescriber notification for failures with potential to cause patient harm (strength, purity, other quality attributes).
- How to determine severity and urgency of the recall event.
- Identifying distribution and all patients who received the affected CNSP.
- Recall of unused dispensed CNSPs; quarantine of all remaining stock.
- Investigation of other potentially affected lots from the same compounding run.
- Disposal and full documentation; reporting to regulatory bodies as required.

Complaint Handling

- Designated person reviews every complaint to determine whether a quality problem exists.
- Investigation considers whether the issue extends to other CNSPs in inventory.
- Written/electronic record retained: complainant, date, nature, response, findings, and follow-up.
- Returned CNSPs are quarantined until destroyed after the investigation closes.

Adverse Event Reporting

- Report per facility SOP and all applicable jurisdictional law requirements.
- If a quality issue is likely to affect other patients, notify those patients and their prescribers promptly.

⚠ A recall plan is only as good as the records it draws from.

Packaging, Transport, and Documentation

Packaging Requirements

- Must preserve physical and chemical integrity and stability of the CNSP throughout its BUD.
- Must protect the CNSP from damage, leakage, contamination, and degradation.
- Must protect compounding and dispensing personnel from exposure to the preparation.

Transport

- Facility SOP defines mode of transport, special handling requirements, and any temperature monitoring needs during transit.

Documentation — Minimum Required Records

- Personnel training, competency assessments, qualifications, and corrective actions.
- Equipment calibration, verification, and maintenance records.
- COAs and supporting documentation for all components.
- SOPs, MFRs, and CRs — complete and current.
- Release inspection, testing records, and cleaning/sanitizing logs.
- Temperature logs and accommodation records.
- Complaints, adverse events, and corrective actions.
- Annual QA/QC and chemical hazard information reviews.

⚠ **Retain all required records for at least 2 years** after preparation — or longer if jurisdiction requires. If it isn't documented, it didn't happen.

Key Takeaways

1. Five Risks

<795> exists to prevent five specific patient harms — every requirement maps back to one of them.

2. Named Accountability

A named designated person owns compliance: training, SOPs, deviations, and quality review. Accountability is personal and written.

3. MFR + CR Required

Every CNSP requires an MFR (the recipe) AND a CR (the per-batch log). No exceptions, no shortcuts.

4. BUD Defaults

BUDs are driven by water activity, preservation, and stability data — the defaults are 14 / 35 / 90 / 180 days at the 0.60 aw threshold.

5. Chapter Layering

Hazardous drugs always layer in <800>. Sterile compounding moves to <797>. <795> is the nonsterile floor, not the ceiling.

6. Documentation Backbone

Documentation is the backbone of every CNSP operation — all records retained for at least 2 years.

⚠ Master these six points and you can defend any inspection.

Questions & Discussion

Open the floor for case-based questions covering the full scope of this module:

BUD Assignment

Practice applying Table 4 thresholds using aw examples and container types to assign correct BUDs.

MFR / CR Completeness

Review sample records and identify missing required elements from Box 2 and Box 3 of <795>.

Garbing Scenarios

Work through garbing and hand-hygiene scenarios, including when glove replacement is required mid-session.

Cleaning Frequency

Apply Table 1 and Table 2 requirements to real-world shift schedules — determine what must be cleaned and when.

References

- USP-NF General Chapter <795>, Official 1-Nov-2023. DOI: 10.31003/USPNF_M99595_06_01
- <51> Antimicrobial Effectiveness Testing
- <659> Packaging and Storage Requirements
- <797> Pharmaceutical Compounding — Sterile Preparations
- <800> Hazardous Drugs — Handling in Healthcare Settings
- <1112> Application of Water Activity Determination to Nonsterile Pharmaceutical Products
- <1163> Quality Assurance in Pharmaceutical Compounding
- <1178> Good Repackaging Practices
- <1231> Water for Pharmaceutical Purposes